Geometry, Triangles & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Parallel Lines & Transversal Angle Relationships

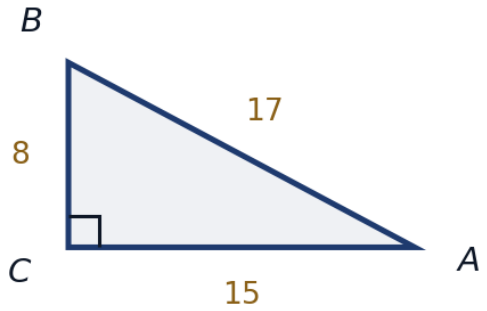
Two parallel lines $\ell_1 \parallel \ell_2$ are intersected by a transversal t. If a pair of consecutive interior (co-interior) angles have measures $(3x + 15)^\circ$ and $(2x - 5)^\circ$, find x and the measure of both angles.

2 [Foundational] • [FRQ] • Target: Triangle Angle-Sum & Exterior Angle Theorem

In $\triangle ABC$, the exterior angle at vertex C measures 134°, and the interior angle $\angle A$ measures 58°. Find $\angle B$ and $\angle ACB$.

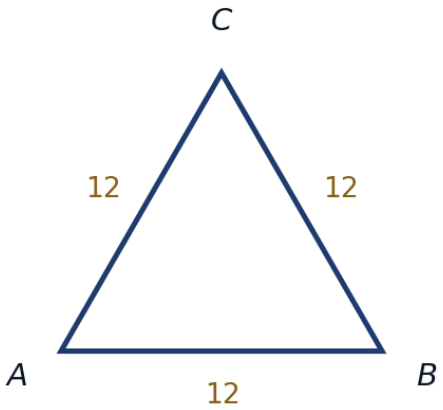
3 [Foundational] • [FRQ] • Target: Pythagorean Theorem & Right-Triangle Trig Ratios

In right triangle ABC shown below, $\angle C = 90^\circ$, $AC = 15$, and $BC = 8$. Find (a) the hypotenuse AB, (b) $\sin A$, (c) $\cos A$, and (d) $\tan B$.



4 [Foundational] • [FRQ] • Target: Special Right Triangles ($30^{\circ}\mathrm{-}60^{\circ}\mathrm{-}90^{\circ}$)

Find the exact perimeter and area of an equilateral triangle whose altitude has length $6\sqrt{3}$ cm.



5 [Foundational] • [FRQ] • Target: Special Right Triangles ($45^{\circ}\mathrm{-}45^{\circ}\mathrm{-}90^{\circ}$)

A square has diagonal of length $10\sqrt{2}$ m. Find its side length, perimeter, and area.

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6 [Foundational] • [FRQ] • Target: Triangle Inequality Theorem

Two sides of a triangle have lengths 7 and 12. State the strict inequality for all possible lengths x of the third side, and find how many integer values x can take.

7 [Foundational] • [FRQ] • Target: Isosceles Triangle Properties

In $\triangle PQR$, $PQ = PR$ (isosceles) and the vertex angle $\angle P$ measures 44°. Find the measure of each base angle $\angle Q$ and $\angle R$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Similar Triangles & Parallel Cross-Sections

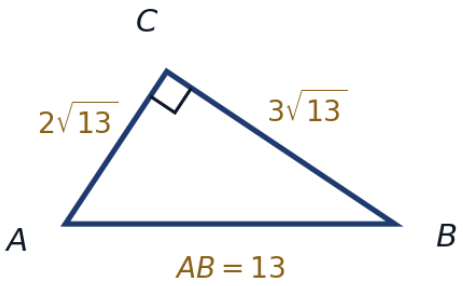
In $\triangle ABC$, segment DE is parallel to base BC with D on AB and E on AC. Given $AD = 6$, $DB = 9$, and $DE = 8$, find the length of BC and the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$.

9 [Intermediate] • [FRQ] • Target: Complementary Angle Identity $\sin\theta = \cos(90^{\circ}-\theta)$

In right triangle DEF with $\angle F = 90^\circ$, $\sin D = \frac{5}{13}$. Find $\cos E$, $\cos D$, and $\tan E$.

10 [Intermediate] • [FRQ] • Target: Geometric Mean Theorems in Right Triangles

In right triangle ABC, the altitude CD is drawn to the hypotenuse AB. If $AD = 4$ and $DB = 9$, find the length of the altitude CD and both legs AC and BC.



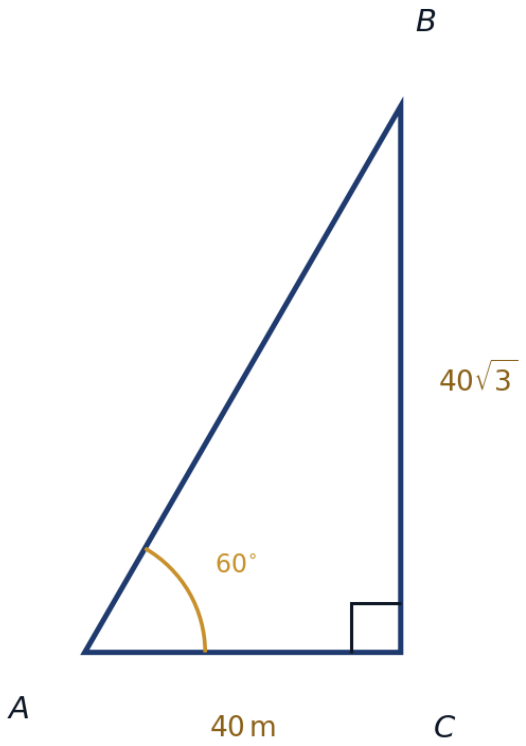
11 [Intermediate] • [FRQ] • Target: Interior & Exterior Angles of Regular Polygons

Find the measure of one interior angle and one exterior angle of a regular 15-sided polygon (pentadecagon).

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12 [Intermediate] • [FRQ] • Target: Angles of Elevation & Depression

A surveyor stands $40\,\text{m}$ from the base of a vertical radio tower on level ground. The angle of elevation to the top of the tower is 60°, and the surveyor's eye level is $1.6\,\text{m}$ above the ground. Find the exact height of the tower and its decimal approximation to two decimal places.

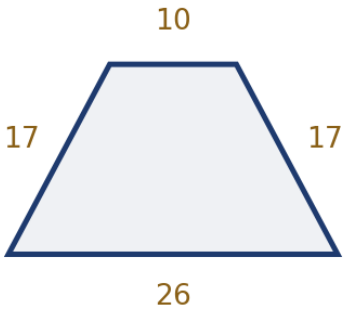


13 [Intermediate] • [FRQ] • Target: Triangle Angle Bisector Theorem

In $\triangle ABC$, the angle bisector of $\angle A$ meets side BC at point D. If $AB = 10$, $AC = 15$, and $BC = 20$, find the exact lengths of BD and DC.

14 [Intermediate] • [FRQ] • Target: Isosceles Trapezoids & Pythagorean Decomposition

An isosceles trapezoid has parallel bases of lengths $10\,\text{cm}$ and $26\,\text{cm}$ and non-parallel legs of length $17\,\text{cm}$. Find the height h, the area, and the length of each diagonal of the trapezoid.



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Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • [FRQ] • Target: Coordinate Triangle Geometry & Centroid • Cross-Lesson: Topic 1.01 (Coordinate Geometry)**

In $\triangle ABC$, the vertices are $A(1, 2)$, $B(9, 2)$, and $C(5, 8)$. Verify that $\triangle ABC$ is isosceles, find the length of the altitude from C to AB, and determine the coordinates of the centroid G.

16 **[Challenging] • [FRQ] • Target: Algebraic Right Triangles via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)**

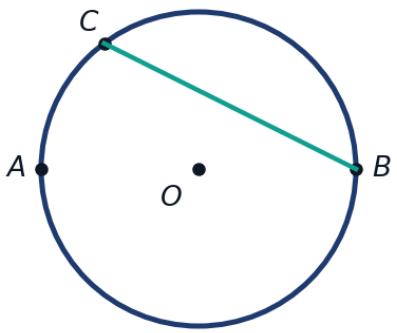
In a right triangle, one leg is $7\,\text{cm}$ longer than the other leg, and the hypotenuse is $17\,\text{cm}$. Set up and solve a quadratic equation to find the lengths of the two legs and the sine of the smaller acute angle.

17 **[Challenging] • [FRQ] • Target: Law of Cosines & Trigonometric Triangle Area • Cross-Lesson: Topic 9.01 (Trigonometry)**

In $\triangle ABC$, $AB = 7$, $BC = 8$, and $\angle B = 60^\circ$. Use the Law of Cosines to find the exact length of side AC and the exact area of $\triangle ABC$.

18 **[Challenging] • [FRQ] • Target: Thales' Theorem & Inscribed Right Triangles • Cross-Lesson: Topic 9.01 (Circles & Inscribed Angles)**

Triangle ABC is inscribed in a circle of diameter $AB = 20\,\text{cm}$. If $\cos A = \frac{3}{5}$, find the lengths of chords AC and BC and the area of the region inside the circle but outside $\triangle ABC$.



19 **[Challenging] • [FRQ] • Target: Inscribed Square via Similar Right Triangles • Cross-Lesson: Topic 5.01 (Proportions) & Topic 8.01 (Area)**

A square is inscribed inside a right triangle with legs 6 and 12 so that one vertex of the square coincides with the right-angle vertex of the triangle and the opposite vertex lies on the hypotenuse. Find the side length s of the square.

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20 [Challenging] • [FRQ] • Target: Orthogonal Medians & Centroid Area Partition • Cross-Lesson: Topic 8.01 (Area Decomposition)

In $\triangle ABC$, medians AD and BE are perpendicular to each other and have lengths $AD = 9$ and $BE = 12$. Using the centroid ratio $2:1$, find the length of side AB and the area of $\triangle ABC$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Triangle Angle-Sum Theorem

In $\triangle ABC$, $\angle A = 42^\circ$ and $\angle B = 73^\circ$. What is the measure of $\angle C$?

a. 55°

b. 65°

c. 75°

d. 115°

22 [Foundational] • [MCQ] • Target: Pythagorean Triples

In a right triangle, the legs have lengths 9 and 12. What is the length of the hypotenuse?

a. 13

b. 15

c. 18

d. 21

23 [Foundational] • [MCQ] • Target: Complementary Trig Identities

If $\sin(3x - 10)^\circ = \cos(2x + 25)^\circ$, where both angles are acute, what is the value of x?

a. 13

b. 15

c. 17

d. 21

24 [Foundational] • [MCQ] • Target: Polygon Interior Angle Sum

What is the sum of the interior angles of a convex octagon (8 sides)?

a. 900°

b. 1080°

c. 1260°

d. 1440°

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25 [Foundational] • [MCQ] • Target: Proportions in Similar Triangles

Triangles ABC and DEF are similar, with AB corresponding to DE. If $AB = 6$, $BC = 10$, and $DE = 9$, what is the length of EF?

a. 12

b. 13

c. 15

d. 18

26 [Foundational] • [MCQ] • Target: Special Right Triangle Ratios

In a $30^\circ\text{–}60^\circ\text{–}90^\circ$ right triangle, the side opposite the 30° angle has length 7. What is the length of the side opposite the 60° angle?

a. $7\sqrt{2}$

b. $7\sqrt{3}$

c. 14

d. $\frac{7\sqrt{3}}{2}$

27 [Foundational] • [MCQ] • Target: Alternate Interior & Supplementary Angles

In the figure below, lines ℓ and m are parallel. If $\angle 1 = 128^\circ$, what is the measure of its alternate interior angle's supplement?

a. 38°

b. 52°

c. 62°

d. 128°

28 [Foundational] • [MCQ] • Target: Testing the Triangle Inequality

Which of the following sets of side lengths CANNOT form a triangle?

a. $4, 5, 8$

b. $6, 6, 11$

c. $5, 9, 14$

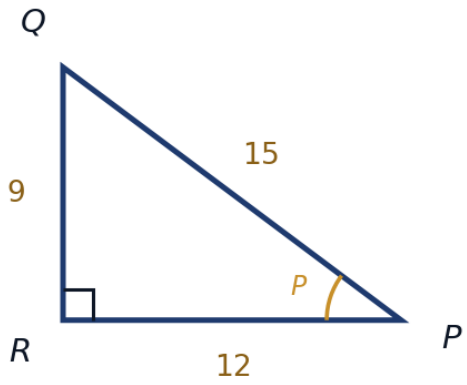
d. $7, 10, 12$

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 **[Intermediate]** • **[MCQ]** • **Target: Perimeter from Trigonometric Ratio**

In right triangle PQR shown below, $\angle R = 90^\circ$, $PR = 12$, and $\tan P = \frac{3}{4}$. What is the perimeter of $\triangle PQR$?



a. 28

b. 32

c. 36

d. 48

30 **[Intermediate]** • **[MCQ]** • **Target: Area-to-Perimeter Ratio in Similar Triangles**

Two similar triangles have perimeters in the ratio $3 : 7$. If the area of the smaller triangle is $27\,\text{cm}^2$, what is the area of the larger triangle?

a. $63\,\text{cm}^2$

b. $108\,\text{cm}^2$

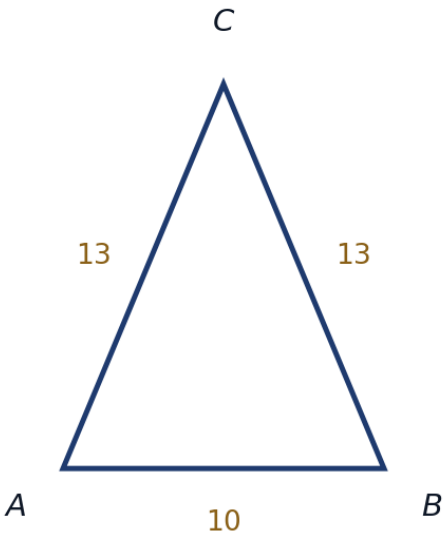
c. $147\,\text{cm}^2$

d. $189\,\text{cm}^2$

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31 **[Intermediate]** • **[MCQ]** • **Target: Altitude & Area of Isosceles Triangles**

In an isosceles triangle with two sides of length 13 and base of length 10, what is the area of the triangle?



a. 60

b. 65

c. 120

d. 130

32 **[Intermediate]** • **[MCQ]** • **Target: Finding Number of Sides from Interior Angle**

If each interior angle of a regular polygon measures 165°, how many sides does the polygon have?

a. 18

b. 20

c. 24

d. 30

33 **[Intermediate]** • **[MCQ]** • **Target: Geometric Mean Theorem for Right-Triangle Legs**

In right triangle ABC with $\angle C = 90^\circ$, altitude CD is drawn to hypotenuse AB. If $AC = 10$ and $AD = 5$, what is the length of AB?

a. 15

b. 20

c. 25

d. $5\sqrt{5}$

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34 [Intermediate] • [MCQ] • Target: Sliding Ladder Pythagorean Word Problems

A 25-foot ladder leans against a vertical wall with its base 7 feet from the wall. If the top of the ladder slides down the wall by 4 feet, how many feet does the base of the ladder slide away from the wall?

a. 4

b. 7

c. 8

d. 15

35 [Intermediate] • [MCQ] • Target: Midsegment Theorem & Area Partition

In $\triangle ABC$, D is the midpoint of AB and E is the midpoint of AC. If the area of trapezoid $DBCE$ is 45, what is the area of $\triangle ABC$?

a. 54

b. 60

c. 75

d. 90

36 [Intermediate] • [MCQ] • Target: Pythagorean Trigonometric Identities

If θ is an acute angle such that $\sin\theta = 3\cos\theta$, what is the value of $\sin\theta\cos\theta$?

a. $\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{3}$

d. $\frac{3}{5}$

37 [Intermediate] • [MCQ] • Target: Altitude Decomposition of Oblique Triangles

In $\triangle ABC$, $\angle A = 30^\circ$, $\angle C = 45^\circ$, and $AB = 12$. What is the exact length of BC?

a. $6\sqrt{2}$

b. $6\sqrt{3}$

c. $6\sqrt{6}$

d. $12\sqrt{2}$

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Inradius of a Right Triangle • Cross-Lesson: Topic 9.01 (Circles & Tangent Segments)

What is the radius of the circle inscribed in a right triangle with legs 9 and 40?

a. 3.5

b. 4

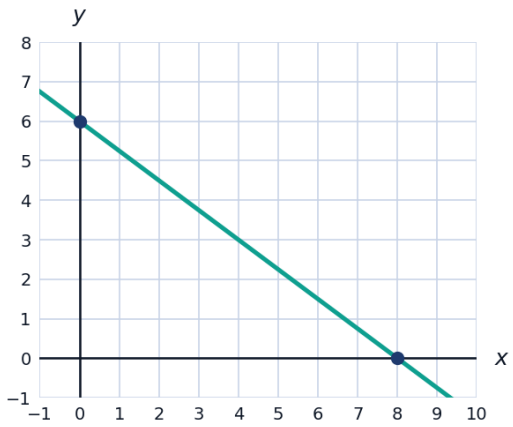
c. 4.5

d. 5

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39 [Challenging] • [MCQ] • Target: Slope as Tangent & Trigonometric Ratios of Lines • Cross-Lesson: Topic 1.01 (Linear Intercepts & Slope)

In the xy-plane, line $y = -\frac{3}{4}x + 6$ forms a right triangle with the positive coordinate axes. What is the value of $\sin\theta$, where θ is the acute angle the line makes with the x-axis?



a. $\frac{3}{5}$

b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

40 [Challenging] • [MCQ] • Target: Heron's Formula & 13-14-15 Triangle Altitude • Cross-Lesson: Topic 8.01 (Area of Triangles)

In $\triangle ABC$, sides a, b, c have lengths $13, 14, 15$. What is the length of the altitude to the side of length 14?

a. 10

b. 11

c. 12

d. 12.5

41 [Challenging] • [MCQ] • Target: Algebraic Squaring with $\sin^2\theta + \cos^2\theta = 1$ • Cross-Lesson: Topic 3.01 (Quadratic Identities)

If $\sin\theta + \cos\theta = \frac{7}{5}$ for an acute angle θ, what is the value of $\sin\theta\cos\theta$?

a. $\frac{12}{25}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{12}{5}$

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42 [Challenging] • [MCQ] • Target: Isoperimetric Area Ratio of Regular Polygons • Cross-Lesson: Topic 5.01 (Ratios) & Topic 8.01 (Polygon Area)

A regular hexagon and an equilateral triangle have the exact same perimeter. What is the ratio of the area of the regular hexagon to the area of the equilateral triangle?

a. $1:1$

b. $3:2$

c. $2:1$

d. $4:3$

43 [Challenging] • [MCQ] • Target: Incenter Angle Formula • Cross-Lesson: Topic 2.01 (Linear Angle Systems)

In $\triangle ABC$, angle bisectors of $\angle B$ and $\angle C$ intersect at the incenter I. If $\angle A = 68^\circ$, what is the measure of $\angle BIC$?

a. 112°

b. 124°

c. 134°

d. 146°

44 [Challenging] • [MCQ] • Target: Crossed-Ladders Similar Triangles Theorem • Cross-Lesson: Topic 10.01 (Harmonic & Rational Equations)

Two poles of heights $10\,\text{m}$ and $15\,\text{m}$ stand vertically on level ground $20\,\text{m}$ apart. A taut wire connects the top of each pole to the base of the opposite pole. At what height above the ground do the two wires intersect?

a. $5\,\text{m}$

b. $6\,\text{m}$

c. $6.25\,\text{m}$

d. $7.5\,\text{m}$

45 [Challenging] • [MCQ] • Target: Coordinate Triangle Area via Bounding Box / Shoelace • Cross-Lesson: Topic 1.01 (Coordinate Plane)

What is the area of the triangle with vertices $(0, 0)$, $(6, 2)$, and $(3, 8)$ in the xy-plane?

a. 18

b. 21

c. 24

d. 27

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